

# research

## Phase 0 Research — Workshop Curriculum Platform

**Feature:** specs/001-workshop-curriculum-platform | **Date:** 2026-09-01

Consolidated from three research streams. Each stream's full evidence — commands, real output, and its own UNVERIFIED register — is in research/:

| Stream              | File                                        | Lines |
|---------------------|---------------------------------------------|-------|
| Transcription       | <u>research/transcription.md</u>            | 1,249 |
| Search architecture | <u>research/search-architecture.md</u>      | 837   |
| LLM bridging        | <u>research/llm-bridging.md</u>             | 735   |
| Reference recon     | <u>../docs/workshop-curriculum/RECON.md</u> | 1,544 |

### Headline: four assumptions the feature request rests on are false

Stated first because every downstream decision follows from them.

1. **ai\_interviewing is not containerised.** Zero container definitions. It runs as a native Go binary (platform/bin/aicur) serving an Angular frontend. “Run it through the containers same way” describes something that does not exist.

2. **No speech-recognition engine is installed.** `/usr/bin/whisper` is `whisper-1.3.1-alt1`, a microphone-loopback GUI by a different author. `import whisper` fails.
  3. **No generative model exists.** `ollama` serves exactly two models and they are the *same* embedding model (`jina-bert-v2`, 160.28M, an encoder with no LM head). The entire model store is 309 MB — too small to hold a decoder. User Story 4 cannot run today under any design.
  4. **Lumen cannot index media, and its chunk IDs are content-derived.** Both are disqualifying for requirements the spec states literally. See D-SEARCH-1 and D-SEARCH-2.
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## D-TRANS-1 — Speech recognition engine

**Decision:** `faster-whisper` (CTranslate2 backend), CPU int8, model `large-v3-turbo`, installed into a project-local virtual environment.

**Rationale:** it is the only candidate that returns `avg_logprob`, `no_speech_prob`, `compression_ratio` and per-word probability. FR-003 requires unclear speech to be *marked* rather than guessed; without a confidence signal that requirement has nothing to key on and becomes editorial opinion. CTranslate2 also avoids the ~900 MB PyTorch wheel.

**Alternatives considered:** - `ollama /v1/audio/transcriptions` — the endpoint is real (verified: it validates content-type, then model presence). But the binary's Whisper strings are an **encoder** projecting audio into an LLM token space (`AudioProjector`, `<|audio_bos|>`); `word_timestamps`, `avg_logprob` and `no_speech_prob` appear nowhere in it. No timestamps kills FR-002/SC-003. Rejected on evidence, not assumption — “assume it can't” would have been wrong about the endpoint existing. - `whisper.cpp 1.9.1` — retained as fallback. Its ALT packaging hard-depends on the CUDA and Vulkan backends, on a machine where this repo deliberately excluded the GPU after an i915 fault silently corrupted embeddings. - *OpenAI hosted API* — rejected by decision D1 (local-only).

**Measured:** recording is **01:55:28.75 (6928.75 s)**, H.264 1080p24, AAC-LC 48 kHz. The “stereo” is **dual-mono** (L-R difference -90.3 dB), so channel-based speaker separation is impossible. Uniformly AGC-compressed. 8 long silences totalling 41.33 s (0.597%).

**Estimate:** 40–80 minutes idle — **UNVERIFIED**, and roughly doubled under current load. Settled by a 5-minute calibration on an already-extracted 300 s sample.

## **D-TRANS-2 — Speaker attribution is human, not automatic**

**Decision:** attribute speakers by human review. Do not run diarization.

**Rationale:** both cues that make diarization tractable are *measurably absent* — no channel separation (dual-mono) and AGC has flattened loudness differences. pyannote would reintroduce the PyTorch dependency D-TRANS-1 avoids, plus a gated model token, to produce a result the audio does not support. FR-005 explicitly permits stating inability.

## **D-TRANS-3 — Accuracy is measured on the audio timeline, not on passages**

**Decision:** word error rate against a blind human reference over  $\geq 30$  seeded, stratified 30-second audio windows.

**Rationale:** this is the load-bearing choice in the whole measurement design. Sampling *machine passages* makes whole-region deletions structurally invisible — a dropped span produces no passage to sample, so accuracy is biased upward precisely where the transcript is worst. Sampling the timeline cannot miss them. Thirty 30 s windows span well over 30 passages, so SC-002's wording is satisfied on its own terms while the bias is removed.

Normaliser hashed and frozen before first measurement. Companion metrics: coverage-gap rate (SC-001), timestamp error median/p95 against the 5 s bound (SC-003), speaker accuracy (FR-005). Honest cost: 1–2 hours of human transcription.

## D-TRANS-4 — Coverage is provable by construction

**Decision:** chunk at  $\leq 300$  s cut *inside measured silence*, condition\_on\_previous\_text=False, atomic checkpoint per chunk, resume by deterministic re-partition plus hash match.

**Rationale:** passage spans  $\cup$  VAD non-speech spans must equal  $[0, 6928.75)$  exactly — so SC-001 becomes an arithmetic identity rather than a judgement. The same setting that makes it resumable (condition\_on\_previous\_text=False) also suppresses Whisper’s repetition-loop hallucinations. The 8 measured silences are a free VAD test fixture.

FR-007 is already implemented by workshop/scripts/extract-videos.sh, which verifies per-part, archive and extracted-video hashes. The pipeline **invokes** it; it does not reimplement it.

## D-TRANS-5 — The notes PDF is a cross-check, never ground truth

**Decision:** use the PDF for section structure, a proper-noun gazetteer, and coverage cross-checking only.

**Rationale:** it has a real text layer (3,034 words — no OCR needed, so tesseract is not on the path) but it is a **summary**: ~26 words per minute of recording. It demonstrably errs, rendering “Spatkit” for SpecKit and “Llama CPP” for llama.cpp. That is concrete evidence for the spec’s rule that the recording wins.

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## D-SEARCH-1 — Lumen chunk IDs cannot carry passage identity

**Decision:** build a passage-identity (PID) layer. Do **not** cite Lumen chunk IDs.

**Rationale** — measured, by controlled three-run re-index:

| Change                                                     | Observed                                       |
|------------------------------------------------------------|------------------------------------------------|
| Fix one typo inside a section, line numbers unchanged      | ID changed 4b98e295c112a429 → 610e608a50f273a6 |
| Prepend a section, shifting every later section by 4 lines | <b>all IDs unchanged</b> ; line ranges moved   |

The IDs are **content-derived and position-independent** — precisely one of the two forms FR-037 forbids, and the one that breaks on the operation this feature performs most often (FR-038 human corrections). Built on them, **SC-016 would score 0% for every corrected passage, silently** — links keep rendering, they just point elsewhere. The IDs are not exposed through the CLI or MCP anyway, leaving only a positional key, which run 3 shows moves. Both available keys fail, in opposite directions.

**Design:** ULIDs minted once at ingest, written back into the source artifact as `<!-- pid: ... -->` anchors, plus a committed registry (`passages.jsonl` → `passages.db`) carrying a `content_hash` column *named so nobody mistakes it for identity*. Citations, cross-references and redactions all resolve through the registry. Code passages get honestly weaker identity (symbol-path keyed with a rename alias table), and that limit is documented rather than hidden.

**Alternatives considered:** reuse Lumen IDs (rejected on the measurement above); positional keys (rejected by run 3); content hashes (identical to the failing case).

## D-SEARCH-2 — Two search paths, not one

**Decision:** lexical FTS5 prefix index for type-ahead; semantic embedding search on submit.

**Rationale — measured under load:** a query embedding takes **18.2–21.0 s**, plus a fixed ~2,202 ms indexer setup per CLI process. An FTS5 prefix index over the 58,726 real symbols builds in 0.93 s / 15.9 MB and answers at **p50 0.25 ms, p95 9.58 ms, p99 19.2 ms**. SC-005's 200 ms budget is unreachable semantically and comfortable lexically, leaving ~190 ms for HTTP and paint. FTS5 works in `modernnc.org/sqlite`, already in `ai_interviewing's go.mod` — no new dependency.

Suggestions therefore return *navigational targets*; submit fuses lexical and semantic.

## D-SEARCH-3 — Media is indexed via text reduction, or not claimed

**Decision:** index transcripts, documentation, code and diagram *descriptions*. Audio and video are searchable **through their transcripts and captions**, not directly.

**Rationale:** Lumen's extension allowlist is a compile-time var with no override — 25 code and doc extensions, and **no** image, audio, video, PDF, HTML or SVG. Measured: zero indexed files for `.sh .html .css .txt .svg .mmd .puml .vtt .srt .pdf`, against 55 tracked `.sh`, 146 `.html` and 32 `.css` in the root module. That is exclusion, not absence. The spec's "all content indexed" is therefore satisfied by reduction to text, and the boundary is stated in the docs (FR-031) rather than implied away.

## D-SEARCH-4 — Health checks must not gate degradation

**Decision:** derive the three-state contract from the *search call*'s own result, never from a health probe.

**Rationale — measured in one MCP session:** `health_check` returned `Status: OK / service is healthy in 2 ms`, and seconds later `semantic_search` failed with `all embedding servers exhausted ... context deadline exceeded`. It is a liveness probe, not a saturation probe. Separately, Lumen returns "No results found. | Warning: Index is being updated in the background..." as one unstructured string — the service must parse that and promote it to Degraded, never forward it as "no results". The anti-pattern already ships in this repo: `ai_interviewing/.../search.component.ts` renders backend errors as the empty state.

## D-SEARCH-5 — Generations and atomic swap

**Decision:** index generations with a verification gate and atomic swap; ingest holds an exclusive lock; reserved embedding capacity for interactive queries.

**Rationale:** measured during the live rebuild, chunks moved 58,734 → 58,744 while `last_indexed_at/root_hash` still advertised the previous generation — a half-written, readable index. Reserved capacity is a hard requirement, not a nicety: SC-006 fails on every chapter ingest without it.

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## D-LLM-1 — Provider seam with four adapters

**Decision:** a three-method Provider interface with adapters `none` (default), `extractive`, `ollama`, `openai_compatible`. Config resolution mirrors `scripts/lumen-reindex.sh` (`env` → `config` → `live probe` → `fallback`), the repo's existing precedent. Defaults enabled: `false`.

**Rationale:** `openai_compatible` covers LM Studio, HelixLLM (its own brief says OpenAI-compatible, and `SSL_CERT_FILE` already trusts its cert) and every external vendor — so FR-023 costs two adapters, not many. Defaulting to disabled means a fresh clone can neither leak nor bluff. FR-025 is enforced by **separate route trees** (compile-time), not a runtime check.

**Alternatives considered:** depend on `LLMProvider` (`git@github.com:HelixDevelopment/LLMProvider`), which genuinely exists at `.../Projects/helix_translate/LLMProvider` with 43 adapters and exactly the interface shape wanted — but its module path `digital.vasic.llmprovider` is unresolvable and it carries `replace digital.vasic.models => ../Models`, a relative-path dependency on a sibling checkout absent from `.gitmodules`. **Copy the interface shape, not the dependency.**

**Trap recorded:** `~/helix-agent` is a 23.7 MB Go binary that is **not** an LLM agent — `--help` shows `-bind-addr` (SWIM), `-wg-key` (WireGuard), `-etcd-endpoints`. It is a cluster mesh node. HelixLLM appears in this repository only as portfolio marketing copy.

## D-LLM-2 — Four-layer grounding, refusal by default

**Decision:** L1 calibrated retrieval gate (`min_score` **and** `min_margin`); L2 JSON-schema- constrained generation where `"minItems": 1` on citations makes an uncited claim structurally undecodable; L3

deterministic citation-ID set membership against the PID registry; L4 support verification (embedding floor, then batched entailment). Any layer failing  $\Rightarrow$  refuse the whole answer; never strip claims silently.

**Rationale:** the margin test is what catches the near-miss case (“what Docker version did he say?”) that scores high on similarity while being unanswerable. **SC-009 is unreachable without L3** — attaching a citation is easy, proving it points at a real passage requires a deterministic set check, which costs microseconds. The design deliberately inverts the usual trust relationship: deterministic checks gate a probabilistic generator.

**Depends on D-SEARCH-1:** L3 requires stable passage IDs. Since Lumen’s are not, the PID layer is a prerequisite for grounding, not an independent nicety.

## D-LLM-3 — SC-010 needs adversarial questions

**Decision:** the  $\geq 10$  unanswerable questions are drawn from a 10-item adversarial taxonomy — near-miss attribute, false premise, uncomputable aggregate, misattributed speaker, lexically-overlapping-but-unanswerable, redacted passage, inaudible segment, and others.

**Rationale:** ten astrophysics questions would pass any threshold and prove nothing. Recorded honestly: 10/10 at temperature 0 is *reproducible*, not *generalising*, and must never be paraphrased as “never fabricates”.

## D-LLM-4 — Privacy is enforced, not configured

**Decision:** resolved-address allowlist (`net.LookupIP + IsLoopback`, not string matching); egress-denied network namespace with a **negative control** — `curl https://example.com` from inside the namespace **MUST** fail; packet capture asserting zero non-loopback packets across the full 20-answer and 10-refusal runs.

**Rationale:** a config flag is not a guarantee. The negative control is what upgrades “we observed no egress” into “egress was impossible”. Hostname string-matching is not a security boundary, so `locality` is **declared**, never inferred.



## D-LLM-5 — “Instant answers” is not achievable; route the ambition

**Decision:** answering is asynchronous. The extractive adapter is the default *working* path.

**Rationale — measured:** three identical two-word embed calls, model resident, minutes apart: **20.16 s / 11.05 s / 0.10 s** at load 8.25. That 200× spread is queue wait, not compute variance — ollama serves one queue and generation contends directly with indexing. Estimated CPU-only generation (UNVERIFIED, method shown): ~21 s (1.5B) / ~42 s (3B) / ~95 s (7B) **idle**, ×1.4 with verification. So “instant” is off by two orders of magnitude and no prompt engineering closes it.

Where the ambition is reachable: search and autocomplete (SC-005/SC-006 are comfortable). The extractive adapter answers in ~0.3 s, is genuinely grounded, **cannot fabricate**, and works today with zero generative capability.

Enabling the GPU is rejected outright — this repository already refused that bargain once, after Vulkan silently corrupted the index.

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## Open items carried into Phase 1

| #  | Item                                                                     | Settles by                                                    |
|----|--------------------------------------------------------------------------|---------------------------------------------------------------|
| U1 | cp314 wheels for CTranslate2/faster-whisper install and run on this host | attempting the venv install                                   |
| U2 | Real transcription throughput on this CPU                                | 5-minute calibration on the extracted 300 s sample            |
| U3 | The recording contains intelligible speech in a known language           | same calibration run — nobody has listened and no ASR has run |
| U4 | Idle embedding latency                                                   | requires the rebuild to finish                                |
| U5 | SC-007/SC-008 retrieval quality                                          | no corpus exists yet                                          |

U1–U3 all close in the same short calibration, which is the single recommended next action.

## Constitution constraints confirmed for Phase 1

- **§11.4.156** — no CI may be added, including inside workshop/. Note: the umbrella’s pre-push gate E was blind to submodules until fixed on 2026-09-01; it now derives the owned fleet from `helix-deps.yaml` and is mutation-proven against two submodules.
- **§11.4.76** requires `vasic-digital/containers` as the sole orchestration layer. It is **not** a submodule of this tree. Container work must resolve this before adding a bespoke stack.  
**WITHDRAWN 2026-09-01 — not silently replaced.** *What was believed when this line was written (2026-08-31): the tree declared no `vasic-digital/containers` gitlink, so the §11.4.76(2) submodule obligation was an unmet prerequisite and any container work was blocked behind an operator decision. What is measured now (2026-09-01): `git config -f .gitmodules --get-regexp containers` returns `submodule.submodules/containers.path submodules/containers` and `submodule.submodules/containers.url git@github.com:vasic-digital/containers.git; submodules/containers` is populated and the gitlink is pinned at `4dab992`. `scripts/verify-governance-cascade.sh` classifies **9** declared submodules — 7 owned (containers among them), 1 governance source, 1 third-party. When it changed: during this feature’s own Phase 1 work, after this line was written and before this correction. The prerequisite is therefore **satisfied, not pending**, and the architecture is settled: containerised, consuming `submodules/containers`. A bespoke stack remains forbidden by §11.4.76(4) — that half of the original sentence was, and is, correct.*
- **§11.4.161** mandates rootless podman. Host has podman 5.7.1 rootless; docker is absent.
- **§1.1** — every gate needs a paired mutation proof that includes a real end-to-end run.